Restricting the semantics of interrogative disjunctors

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Poster and references

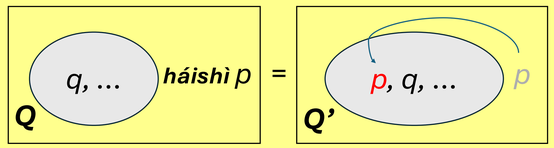
Introduction

- In some languages, forming alternative questions (AltQs) requires a special disjunction marker (Haspelmath 2007, Mauri 2008).
- In Mandarin, one type of AltQs is formed with a specific morpheme ***háishì.***

(1) Ā Máo xǐhuān hē kāfēi (ne,) {*huòzhě/ háishì} xǐhuān hē chá (ne)?
A Mao like drink coffee NE or or like drink tea NE
'Which one is true: A Mao likes drinking coffee or tea?' (AltQ)

- Previous analyses all share that ***háishì*** is a **symmetric** connective (Huang 1991, Erlewine 2025, a.o.)
 - The syntactic-semantic properties of the disjuncts are identical.
- In this paper, I argue **against this symmetry**:
 - Instead, [S_L ***háishì*** S_R] expresses an asymmetric mapping from a proposition *p* and a question *Q* to form a new question *Q'*, roughly "***Q or p***?".

(2) [S_L ***háishì*** S_R]



- The proposition, ***p***, is added to the set of possible answers for ***Q***.
- If ***Q*** is a Polar Q, then its positive answer ***q*** forms an AltQ with ***p***: *"**q or p**?"*

- Two empirical asymmetries in *háishì*-constructions:
 - (a) the distribution of **polar question particles**
 - Only the right disjunct of *háishì* resists *ma*-marked.
 - (b) the acceptability of *wh*-questions
 - Only the right disjunct of *háishì* cannot be a *wh*-question.

Challenge 1: PolQ *particle*

Observe the distribution of **polar question particles:**
- Only the right disjunct (S_R) of *háishì* resists *ma*-marked.

(8) *Ā Máo xǐhuān hē kāfēi {**ma**/ ne/ ∅}, háishì xǐhuān hē chá **ma**? ***[… *háishì … ma*]**
A Mao like drink coffee QP NE or like drink tea QP
(Judgment from Mainland Mandarin speakers; cf. Taiwan Mandarin, Hsieh 2026)

(9) Ā Máo xǐhuān hē kāfēi {**ma**/ ne/ ∅}, háishì xǐhuān hē chá (ne)? **✓[… *háishì* … (*ne*)]**
A Mao like drink coffee QP NE or like drink tea NE
'Does A Mao like drinking coffee, or does he like drinking tea?'

- What is the function of *ma*?
 - *Ma* occurs only with matrix PolQ (not other kinds of questions).

(10) Ā Máo xǐhuān hē kāfēi **ma**?
A Mao like drink coffee QP
'Does A Mao like drinking coffee?'

(11) Ā Máo xǐhuān hē shénme **ma**?
A Mao like drink what QP
#'What does A Mao like drinking?'

→ It is thus commonly understood as marking a PolQ (Cheng 1991, Paul 2014, Pan 2019, a.o.)

→ This rules out the possibility that *ma* scopes above *háishì* yielding an AltQ reading.

*[[$_{\text{AltQ}}$ S_L … *háishì* … S_R] *ma*]

→ But S_L can be *ma*-marked, it would be possible for S_R to be *ma*-marked as well, assuming symmetry.

*[$_{\text{AltQ}}$ [S_L …] *háishì* [[… S_R] *ma*]]

➔ I suggest the anomaly can be explained if S_R cannot be a PolQ. What it is in (9) is a proposition.

Challenge 2: *wh-question*

Observe the acceptability of *wh*-questions
- Only the right disjunct (S_R) of *háishì* cannot be a *wh*-question.

(12) Nǐ de diànhuà shì **duōshǎo**, {huòzhě /*háishì} nǐ de yóuxiāng shì **shénme**? **[*wh*Q+*wh*Q]**
2SG DE phone number be how many or or you DE email be what
'(To recover the username, please tell me) what is your phone number, or what is your email?'

(13) Nǐ dǎsuàn zài **nǎr** zū chē (ne), háishì dǎsuàn jiè yí-liàng (ne)? **[*wh*Q + *háishì*(p)]**
2SG plan LOC **where** rent car NE or plan borrow one-CL NE
'Where do you plan to rent a car, or do you plan to borrow one?' (after Ciardelli et al., 2018)

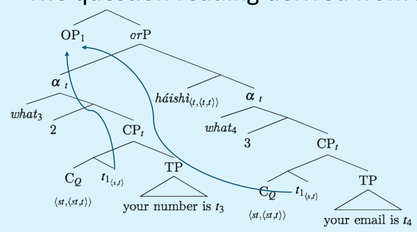
- Using a sequence of two *wh*-questions is bad.
- But when the right clause is arguably not a question, the sequence is acceptable.

Erlewine (2025): arguments of *háishì* focus alternatives.
→ $[\![p]\!]^{alt} = \{p\}$ Both disjuncts are of the type <st,t>,
$[\![Q]\!]^{alt} = \{p_1, p_2, \ldots\}$ no semantic explanation for (8) and (12).

Huang (1991): *háishì* expresses a Boolean V.

*LF: [**Op**$_1$ [*what*$_3$ [[C$_Q$ **t**$_1$] your-number-is t$_3$] *háishì*$_{\langle t,\langle t,t\rangle\rangle}$ [*what*$_4$ [[C$_Q$ **t**$_1$] your-email-is t$_4$]]]]

→ The Boolean V would make available such an LF (vF&H 2021).
The question reading derived from ATB-movement of *Op.*



→ Disjoining *wh*-questions would be predicted possible.

➔ Again, if S_R cannot be a question, (12) is predicted.

Background and previous analyses

(1) Ā Máo xǐhuān hē kāfēi (ne,) {*huòzhě/ háishì} xǐhuān hē chá (ne)?
A Mao like drink coffee NE or or like drink tea NE
'Which one is true: A Mao likes drinking coffee or tea?' (AltQ)

- In (1), *háishì* connects two seemingly declarative clauses S_L and S_R.
 - Neither is marked by any overt interrogative morpheme (*ne* is not one).
 - But the disjuncts might as well be a question:

Ā Máo xǐhuān hē kāfēi is ambiguous btn declarative and interrogative.

- H1: S_L and S_R denotes propositions, and the combination expresses an AltQ with those two propositions as its possible answers (Li & Thompson 1981).

(3) $[\![\textit{háishì}]\!] = \lambda p_{st}.\lambda q_{st}.\lambda r_{st}.\ r = p \vee r = q.$

- H2: S_L and S_R denotes propositions, and *háishì* express the Boolean operator ∨, the AltQ derived via scoping the disjunction over an interrogative C (Huang 1991, after Karttunen 1977, von Fintel & Heim 2021): **[[p or q]** C$_Q$ [… ~~[p or q]~~ …]]

(4) $[\![\textit{háishì}]\!] = \lambda p_t.\lambda q_t.\ p \vee q.$

- H3: *háishì* expresses set union ∪ in the domain of Focus Semantic Value (Erlewine 2025, after Rooth 1992, Beck 2006). The denotation of *háishì* is compatible with disjuncts of varying semantic types (<st,t>, <s,t>, e, etc.).

(5) $[\![\beta\ \textit{háishì}\ \alpha]\!]^{alt} = [\![\beta]\!]^{alt} \cup [\![\alpha]\!]^{alt}$

I will assume the following LF for *háishì*:

(6) ***LF:*** [β [*háishì* α]]

H1 is challenged by the recursive uses of *háishì*:
- The output of one *háishì* doesn't feed the composition with the other.

(7) Ā Máo jīntiān **(háishì)** míngtiān **háishì** hòutiān lái (ne)?
A Mao today or tomorrow or in 2 days come NE
'Will A Mao come today, tomorrow, or in two days?'

Given the input and output of *háishì* are of the same type, H2 and H3 easily allow recursion.

!! Semantically, all these analyses assume that the disjuncts connected by *háishì* are symmetric.

Proposal and discussions

Proposal: The observed semantic asymmetries are reflected in the semantics of *háishì.*

(14) $[\![\textit{háishì}]\!]^w := \lambda p_{st}.\ \lambda Q_{stt}.\ \lambda r_{st}.\ [Q(r) \vee r = p]$

→ S_L must be a question, and S_R must be a proposition..
→ When S_L is a PolQ, a monopolar analysis is assumed: *Q* denotes a singleton set {*p*}. (Roberts 1996, Uegaki 2014).

- One might wonder if *háishì* does not scope over the preceding questions in (8) and (12):

Q1: [S_L]? Q2 : [*Háishì* S_R]?

→ To maintain the semantics as in (14), one can postulate a pronoun anaphoric to the salient preceding question.

Q1: [S_L]? Q2 : [***pro***$_{\mathbf{Q1}}$ *Háishì* S_R]?

Consequences of the semantics in (14):
- No movement assumed → island insensitivity of *háishì* AltQs follows naturally.
- For clause-medial *háishì* AltQs in Mandarin, I need to assume they all involve ellipsis from biclausals.
- Another disjunctor *huòzhě* **never** forms AltQs in Mandarin (1), its semantics also needing restriction to block these constructions altogether.